

# Day 1: DSA Prerequisites

Basics of C++ before Non-Linear Data Structures (Trees, Graphs, Heaps)

Before starting Non-Linear Data Structures (Trees, Graphs, Heaps), you should become comfortable with these five basic programming concepts:

- Variables and data types
- Basic input/output (cin and cout)
- Conditional statements (if, if-else, nested if)
- for loop
- while loop

Each concept is explained in simple language, with when to apply it and three practice programs with user-defined input. All examples use basic C++ and can be compiled in Code::Blocks, Dev-C++, Visual Studio, or an online C++ compiler.

## 1. Variables and Data Types

### 1.1 What is a variable?

A variable is a named memory location used to store a value that can be used or changed during program execution. Think of a variable as a labelled box.

**Example: Student marks** | Variable name: marks | Stored value: 85

```
int marks = 85;
```

Here:

- `int` specifies the data type.
- `marks` is the variable name.
- `85` is the value stored in the variable.
- `=` is the assignment operator.

### 1.2 Common data types in C++

| Data type           | Stores                             | Example                            |
|---------------------|------------------------------------|------------------------------------|
| <code>int</code>    | Whole numbers                      | <code>int age = 20;</code>         |
| <code>float</code>  | Decimal values                     | <code>float price = 25.5;</code>   |
| <code>double</code> | Decimal values with more precision | <code>double pi = 3.14159;</code>  |
| <code>char</code>   | One character                      | <code>char grade = 'A';</code>     |
| <code>string</code> | Text                               | <code>string name = "Neha";</code> |
| <code>bool</code>   | True or false                      | <code>bool passed = true;</code>   |

For `string`, include the header `#include <string>`. The examples below use `#include <iostream>` and, where needed, `#include <string>`.

## 1.3 Basic input and output

- `cout` displays information on the screen.
- `cin` accepts input from the keyboard.
- `>>` takes input into a variable.
- `<<` sends output to the screen.

```
#include <iostream>
using namespace std;

int main()
{
    int age;
    cout << "Enter your age: ";
    cin >> age;
    cout << "Your age is " << age;
    return 0;
}
```

### Sample execution:

```
Enter your age: 21
Your age is 21
```

## When should you use variables and input/output?

Use variables whenever a program needs to remember information, such as:

- Student roll number, marks, and name.
- Number of vertices or edges in a graph.
- Number of nodes in a tree.
- Array elements and search keys.
- Values entered by the user in a menu-driven program.

For example, in a graph program, `int vertices;` stores the number of vertices supplied by the user.

## Practice 1: Variables and Input/Output

These three programs help you understand how to accept different types of input, store them, and display results.

### Example 1: Student details

**Problem:** Accept a student's name, roll number, and marks, then display the details.

```
#include <iostream>
#include <string>
using namespace std;
int main()
{
    string name;
    int rollNo;
    float marks;
    cout << "Enter student name: ";
    cin >> name;
    cout << "Enter roll number: ";
    cin >> rollNo;
    cout << "Enter marks: ";
    cin >> marks;
    cout << "\nStudent Details\n";
    cout << "Name: " << name << endl;
    cout << "Roll No: " << rollNo << endl;
    cout << "Marks: " << marks << endl;

    return 0;
}
```

#### Sample input:

```
Enter student name: Amit
Enter roll number: 12
Enter marks: 87.5
```

#### Output:

```
Student Details
Name: Amit
Roll No: 12
Marks: 87.5
```

## Example 2: Calculate the area of a rectangle

**Problem:** Accept length and breadth from the user and calculate the area.

**Formula:** Area = Length × Breadth

```
#include <iostream>
using namespace std;
int main()
{
    float length, breadth, area;
    cout << "Enter length: ";
    cin >> length;
    cout << "Enter breadth: ";
    cin >> breadth;
    area = length * breadth;
    cout << "Area of rectangle = " << area;
    return 0;
}
```

**Sample input:** length = 5, breadth = 3

#### Output:

Area of rectangle = 15

**Where to apply:** Calculations involving marks, areas, prices, salaries, and measurements.

### Example 3: Swap two numbers

**Problem:** Accept two numbers and exchange their values using a temporary variable.

```
#include <iostream>
using namespace std;
int main()
{
    int a, b, temp;
    cout << "Enter two numbers: ";
    cin >> a >> b;
    temp = a;
    a = b;
    b = temp;
    cout << "After swapping:\n";
    cout << "a = " << a << endl;
    cout << "b = " << b << endl;
    return 0;
}
```

**Sample input:**

Enter two numbers: 10 20

**Output:**

```
After swapping:
a = 20
b = 10
```

**Where to apply:** Swapping values, sorting elements, and exchanging data stored in array positions.